Department of MECHANICAL ENGINEERING

Department of Mechanical Engineering

Clemson University

ME-3040-001

Heat Transfer

Spring 2026

REPORT ON

"Design of the Cooling Mechanism of an Electronic Device"

Submitted by

William Meany

On

April 16, 2026

To

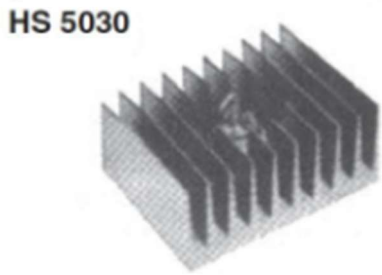
Professor Yuhao Xu

ME 3040-001 Heat Transfer Project, Spring 2026
Design of the Cooling Mechanism of an Electronic Device

Due: April 16, 2026, at 5:00 pm (EDT)
Both hard-copy and soft-copy submissions are required.

PROJECT DESCRIPTION

Consider 100 mm by 40 mm thin rectangular electronic devices assembled in mass quantities at a manufacturing plant. The top surface of the electronic device is made of aluminum and is attached with an HS 5030 aluminum heat sink positioned horizontally. An image of the HS 5030 heat sink is shown on the right. The electronic device dissipates 50.8 W of heat through the heat sink. Both aluminum surfaces have a roughness of about 10 μm and an average interface pressure of 10^5 N/m^2). To prevent the electronic device from overheating, the top surface temperature should be kept below 85°C when the ambient temperature is 23°C. The combined (natural convection and radiation) thermal resistance of an HS 5030 heat sink, attached horizontally, is 1.2 K/W.



Your task as a product engineer is to ensure that the heat sink attached to the device's top surface is capable of keeping the device from overheating. You need to determine whether the top surface temperature of the electronic device with an HS 5030 heat sink attached exceeds 85°C. If so, what action can be taken to reduce the surface temperature to below 85°C? Complete the following tasks for this design project.

(a) Using relevant tables in the required textbook for this class (as outlined in the syllabus), find the thermal contact resistance, in m2·K/W, for aluminum plates with air at the interface (10-μm surface roughness, 10^5 N/m^2). Provide the source (e.g., table number) of your answer.

(b) Calculate the top surface temperature of the electronic device, with an HS 5030 heat sink attached to it, and determine whether this surface temperature is higher than 85°C. Clearly list all the *assumptions* you use for your calculations. Draw a *thermal resistance network* to support your calculations, and clearly label the heat transfer rate (or the heat flux, as appropriate), all temperatures, and corresponding thermal resistances on the network. Show all steps in your calculations and type equations directly in Microsoft Word.

(c) If the surface temperature computed in (b) exceeds 85°C, what action can be taken to reduce the surface temperature to below 85°C? Provide relevant calculations to support your conclusion. Note that, since your company has a large inventory of HS 5030 heat sinks, replacing them with more effective ones is too costly and is not a viable solution. The table you used in (a) may provide some potential directions/solutions.

(d) Perform a literature review to find out the methods used in engineering practices to reduce thermal contact resistance in electronics and thus prevent the electronic devices from overheating. Discuss your findings. Note that you should have a "References" section at the end of your report to include all books and other materials consulted in the process of the literature review. All reference materials should be directly referenced in the report. References should be listed at the end of the report in a consistent format; the ASME standard for referencing is recommended.

ABSTRACT

Electrical components draw current and in return produce heat when being utilized. The buildup of heat can cause electronics to slow their processing, or excessive heat can damage components. The issue of discussion is the surface temperature of a thin electronic device, with an aluminum construction. This device produces a fixed amount of heat, which needs to be dissipated through an aluminum heat sink. The topic of study will be the interface between these two components, and how the efficiency of this process may be improved. Through calculations, it was determined that the surface temperature of the component exceeded safe limits, and therefore a redesign was required. The interfacial fluid during the initial calculations was air, but when replaced with glycerine, the thermal resistance decreases enough to allow safe operations. Further research into other solutions yielded two possibilities. The first is the replacement of the interfacial fluid with a metal alloy that remains a solid at room temperature, before melting into a fluid when undergoing heat transfer. The second is with the interface itself, as an increase in interfacial pressure, or a reduction in hardness will lead to a decrease in the thermal resistance.

TASK A

The thermal contact resistance ($R''_{t,c}$) for an aluminum interface with a surface roughness of 10 μm and an interfacial pressure of 10^5 N/m^2 is found from TABLE 1.

Thermal Resistance, $R''_{t,c} \times 10^4$ ($m^2 \cdot K/W$)				
(*a*) Vacuum Interface			**(*b*) Interfacial Fluid**	
Contact pressure	100 kN/m^2	10,000 kN/m^2	Air	2.75
Stainless steel	6–25	0.7–4.0	Helium	1.05
Copper	1–10	0.1–0.5	Hydrogen	0.720
Magnesium	1.5–3.5	0.2–0.4	Silicone oil	0.525
Aluminum	1.5–5.0	0.2–0.4	Glycerine	0.265

TABLE 1: Thermal contact resistance values for (a) metallic interfaces under vacuum conditions and (b) aluminum interface (10 μm surface roughness, 10^5 N/m^2) with different interfacial fluids. This is TABLE 3.1 [5]

The value of $R''_{t,c}$ is therefore $2.75 \times 10^{-4} m^2 \cdot K/W$

TASK B

In order to calculate the surface temperature of the electronic device with the heat sink attached, a thermal resistance network is created. This network, show below, assumes the conduction is one dimensional, steady state operations, constant properties, negligible thermal resistance of the electrical component, and there is no convection from the electronic component. The thermal resistance diagram is below in FIG. 1.

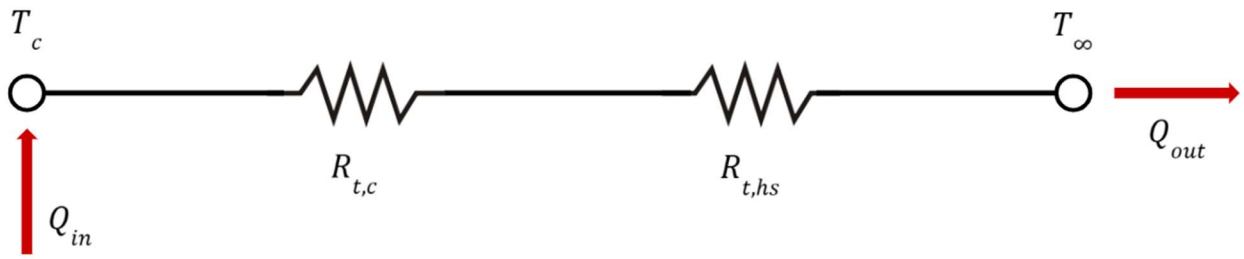


FIGURE 1: Thermal resistance diagram of the system

T_c represents the temperature of the component, which due to the nature of the thin electronic device, is the surface temperature. Q_{in} is the wattage produced, which is 50.4W. $R_{t,c}$ is the thermal resistance of the interfacial fluid, and $R_{t,hs}$ is the thermal resistance of the heat sink. This value includes both the natural convection, as well as the radiation. T_∞ is the temperature of the surrounding air, and Q_{out} is the heat dissipated by the heat sink. $R_{t,c}$ is needed to be solved for using $R''_{t,c}$. This is because the areas are not uniform throughout the interfaces. The area of the interface (A) needs to be calculated, and $R''_{t,c}$ is a value from TABLE 1.

$$A = l * w = 0.1m * 0.04m = 0.004m^2 \quad (1)$$

$$R_{t,c} = \frac{R_{t,c}^{''}}{A} = \frac{2.75 \times 10^{-4} m^2 \cdot K/W}{0.004 m^2} = 0.06875 K/W \quad (2)$$

$R_{t,hs}$ is a given value of 1.2K/W. The energy balance equation can be applied to the system

$$Q_{in} = Q_{out} = \frac{T_c - T_\infty}{R_{t,c} + R_{t,hs}} \quad (3)$$

This equation can be rearranged to solve for T_c, which is the surface temperature. The surrounding temperature can remain in Celsius due to thermal resistance being based on change in temperature.

$$T_c = Q_{in}\left(R_{t,c} + R_{t,hs}\right) + T_\infty \quad (4)$$

$$T_c = 50.4W(0.06875K/W + 1.2K/W) + 23℃ = 87.4525℃$$

This value is larger than the maximum temperature of 85℃ which means that the electronic device will be overheating when operating at a steady state.

TASK C

The surface temperature needs to be reduced such that the steady state operating temperature is below the 85℃ threshold. The heat sink is not to be replaced, and the design of the component is not to be changed. Therefore, the increase in performance must come from the interface between the two. The $R_{t,c}^{''}$ value previously obtained from TABLE 1 was for air in the interface. The table has other options with glycerine having the lowest thermal resistance. This value of $R_{t,g}^{''} = 0.265 \times 10^{-4} m^2 \cdot K/W$ is a reduction of more than a factor of ten. The replacement of the interfacial fluid also allows the surface roughness, and pressure to remain the same. This minimizes the design changes. Recompleting the calculations yields the following results.

$$R_{t,g} = \frac{0.265 \times 10^{-4} m^2 \cdot K/W}{0.004 m^2} = 0.006625 K/W$$

$$T_c = 50.4W(0.006625K/W + 1.2K/W) + 23℃ = 83.8139℃$$

The replacement of air in the interface with gylcerine results is a temperature below the overheating temperature. Glycerine is relatively cost effective, and by replacing the air as the interface fluid, the overheating will be prevented, and no physical changes are required for the heat sink or electrical component.

TASK D

The performance of electronic components is closely tied to their temperature. If an electronic component cannot sustain a safe temperature while operating, the performance will have to be throttled down to keep the component safe. One proposal is the replacement of the interfacial fluid with an alloy [2]. This alloy is designed to melt at the steady state operating temperature, and therefore fills the space between components completely. The advantage of using an alloy is that

the thermal contact resistance is virtually zero [4]. The melting point of specific alloys is around 60℃, so their application with the electrical component is possible. At room temperature, the alloy will be a solid, and therefore the components could be separated for replacement or other forms of servicing. The downsides of this method are the increased costs of an alloy for specific operational temperature windows. This additional cost can be impractical if other solutions achieve the same goals for cheaper, especially for large scale production.

The metal alloy works well due to two factors, the ability to cover the entire surface area of contact, and the low thermal contact resistance of metals. Therefore, if the application of metals is not practical, the low thermal contact resistance properties of must be disregarded. This leaves the interfacing surface area as the interest. A higher contact pressure, as well as lower hardness will allow for more heat transfer to occur [3]. These two changes work by creating micro-contacts which increase the surface area. This increase in surface area, as seen in equation 2, decreases the thermal resistance. The application of higher contact pressure can be achieved through stronger fasteners, and the lower hardness materials can be selected during the design process. The downsides of this method are the small results. When compared to the alloy method mentioned above, the increase of contact pressure and material hardness can reduce the thermal contact resistance, but not make it virtually zero.

Thermal contact resistance is an interesting problem, due to the scale that it affects everyday life. Electrical components surround modern life, and their constant production of heat while operating must be a design consideration when implementing them into everyday devices. The understanding of the core problem allows these design decisions to be informed. Thermal contact resistance serves as one means to increase efficiency, and ensure safe operation long after the component leaves production and enters the world.

References

[1] “Glycerin Livestock,” NOSB Material Database, Jan. 2010. Accessed: Apr. 15, 2026. [Online]. Available: https://www.ams.usda.gov/sites/default/files/media/Glycerine%20TR.pdf

[2] R. S. Cook, K. H. Token, and R. L. Calkins, “A novel concept for reducing thermal contact resistance | Journal of Spacecraft and Rockets,” *Journal of Spacecraft and Rockets*, vol. 21, no. 1, p. 122, 2025, doi: https://doi.org/10.2514/jsr.1984.21.issue-1;journal:journal:jsr;article:article:10.2514/3.8618;website:website:aiaa-site;wgroup:string:AIAA.

[3] M. Grujicic, C. L. Zhao, and E. C. Dusel, “The effect of thermal contact resistance on heat management in the electronic packaging,” *Applied Surface Science*, vol. 246, no. 1–3, pp. 290–302, Jun. 2005, doi: https://doi.org/10.1016/j.apsusc.2004.11.030.\

[4] “Liquid Metal Thermal Interface Materials 1,” *Indium Corporation*. https://www.indium.com/blog/liquid-metal-thermal-interface-materials-1/ (accessed Apr. 16, 2026).

[5] T. L. Bergman, *Introduction to heat transfer*, 6th ed. Wiley, 2011, pp. 118–127.